Status	Finished
Started	Friday, 21 August 2026, 11:19 AM
Completed	Tuesday, 25 August 2026, 3:16 PM
Duration	4 days 3 hours
Grade	**10.00** out of 10.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Given a positive integer a, find the smallest positive integer b whose multiplication of each digit equals to a.

If there is no answer or the answer is not fit in 32-bit signed integer, then return 0.

Example 1

Input:

48

Output:

68

Example 2

Input:

15

Output:

35

For example:

Input	Result
48	68
15	35

Answer: (penalty regime: 0 %)

```
import sys

def smallest_factor_integer(a):
    # Single-digit numbers are their own smallest answer
    if a < 10:
        return a

    digits = []

    # Greedily divide by the largest possible single-digit numbers (9 down to 2)
    for d in range(9, 1, -1):
        while a % d == 0:
            digits.append(str(d))
            a //= d

    # If a could not be reduced to 1, it has prime factors >= 11
    if a > 1:
        return 0
```

	Input	Expected	Got	
✓	48	68	68	✓
✓	15	35	35	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **2**

Correct

Mark 1.00 out of 1.00

Let's print a chessboard!

Write a program that takes input:

Integer N(represents the rows and columns of a chessboard) and also the starting character of the chessboard

Output Format

Print the chessboard as per the given examples

Sample Input / Output

Input:

2

W

Output:

WB

BW

Answer: (penalty regime: 0 %)

```
# Read input: N (integer) and starting character (string)
n = int(input())
start_char = input().strip()

# Determine the alternate character (assuming standard chessboard switching between W and B, or
toggling characters)
# If start_char is 'W', alt_char is 'B'. If start_char is 'B', alt_char is 'W'.
# More generally, we can toggle between two characters.
alt_char = 'B' if start_char == 'W' else 'W'

for i in range(n):
    row = []
    for j in range(n):
        # Alternate characters based on the sum of row and column indices
        if (i + j) % 2 == 0:
            row.append(start_char)
        else:
            row.append(alt_char)
```

	Input	Expected	Got	
✓	2 W	WB BW	WB BW	✓
✓	3 B	BWB WBW BWB	BWB WBW BWB	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **3**

Correct

Mark 1.00 out of 1.00